

When Encoders Hurt: Diagnosing and Stabilising Late-Training Instability in Graph-Augmented Soft Q-Mixing

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Abstract

Graph attention encoders are a routine component of value-decomposition methods for cooperative multi-agent reinforcement learning, where they are presumed to inject inductive biases that help agents share information. We report a previously undocumented failure mode that arises when one such encoder (a two-layer entity / cross-agent graph attention block) is bolted onto Soft-QMIX, a recent state-of-the-art entropy-regularised value-decomposition algorithm. After approximately 6M environment steps on SMACv2 Protoss 5v5, the augmented system stops trying to *win* and instead learns to *survive*: episode length grows by 66%, the number of allies that die per episode falls by 19%, and test-time win-rate regresses from a 0.79 peak to 0.69. We trace the mechanism analytically — the entropy bonus in the Soft-QMIX target inflates with episode length, and asymmetric per-agent uncertainty induced by the encoder turns this inflation into an attractive basin around stalling — and confirm it experimentally with a controlled per-agent- α premise test that produces the same signature in extreme form (test win rate collapses to 0.51). We then propose three lightweight stabilising components: (i) decoupling the per-agent α used in policy sampling from the *uniform* α used in the entropy bonus of the target, (ii) a Q-spread per-agent sensor that is 3.4 $\times$ more discriminative than a previously tried InfoNCE coordination signal, and (iii) a curriculum that fades heterogeneous α in only after a uniform- α scalar warmup. The combined system removes the late-training regression and matches a re-implemented pure Soft-QMIX baseline (last-100-checkpoint test win rate: 0.731 vs 0.729). We do **not** claim improvement over pure Soft-QMIX; we contribute (a) a clean characterisation of an instability mode that practitioners adding encoders to Soft-style methods will likely encounter, (b) a working remediation, and (c) a documented negative result on InfoNCE-based per-agent coordination sensing in role-randomised SMACv2.

1. Introduction

Cooperative multi-agent reinforcement learning (MARL) has converged on a small set of structurally similar building blocks: a per-agent recurrent encoder, a value-decomposition mixer (VDN, QMIX, QPLEX), and a centralised-training / decentralised-execution learner operating on the joint action–value. Recent work has decorated this stack along several axes: stronger encoders (graph attention, transformers, attention-over-entities), stronger objectives (entropy regularisation, contrastive auxiliaries), and explicit per-agent or per-role heterogeneity.

These additions are typically presented as *Pareto-improvements* — better representation, better objective, no harm done. We show this presumption is wrong for at least one combination that is natural to try: a graph attention encoder coupled to Soft-QMIX. The combination *does* learn faster early in training and peaks higher than the unaugmented baseline, but then regresses, ending below the baseline at 10M training steps. The regression has a clean signature — episode length grows, agents stop dying, fewer enemies die — that distinguishes it from generic plateau or overfitting.

We diagnose the mechanism, replicate it in extreme form by deliberately inducing it (a per-agent fixed- α variant collapses to 0.51 win-rate by 10M), and propose three small modifications that remove it. The stabilised system matches the unaugmented baseline; importantly, it does *not* exceed it. We do not claim a state-of-the-art result. We claim a useful *negative* result and a useful *characterisation*: practitioners who add encoders to entropy-regularised value-decomposition methods can expect this failure mode, and the remediation is mechanically simple.

Contributions

1. We document a previously unreported late-training stalling phenomenon in encoder-augmented Soft-QMIX (§3) and identify the entropy-bonus accumulation mechanism (§4) that drives it.
2. We confirm the mechanism via a controlled per-agent- α premise test that reproduces stalling in extreme form (§4.3).
3. We propose three remediation components — decoupled sampling-vs-target α , a Q-spread-based per-agent sensor, and a uniform-to-heterogeneous curriculum (§5) — and show they jointly remove the

regression while leaving asymptotic performance intact (§6).

4. We report a documented negative result: an InfoNCE-based per-agent coordination signal fails to differentiate agents in role-randomised SMACv2 by a factor of 3.4× compared to our simpler Q-spread sensor (§6.4).

2. Background

2.1 Cooperative MARL and value decomposition

We consider Dec-POMDPs $\langle S, A, O, P, R, n, \gamma \rangle$ with n cooperative agents sharing reward r_t . Each agent i holds a partial observation history τ_i and outputs a per-agent action–value $Q_i(\tau_i, \cdot)$. Value-decomposition methods approximate the joint Q_{tot} as a monotone combination of per-agent Q_i , enabling decentralised execution from centralised training.

2.2 Soft-QMIX

Soft-QMIX is a recent entropy-regularised variant. The mixer combines a VDN-like sum with two learned operators: $g(\cdot)$, an order-preserving residual that shapes per-agent Q values without changing their action ranking, and $f(\cdot)$, an affine per-agent operator (a learned per-state temperature).

Action selection during training samples from the soft policy

$$\pi_i(a | s) = \text{softmax}(f_i(g_i(Q_i(s, \cdot))) / \alpha)$$

and the TD(λ) target uses a sample-based entropy bonus,

$$y_t = r_t + \gamma (Q_{\text{tot}}^{\text{tgt}}(s_{t+1}, a_{t+1}^*) - \alpha \cdot \sum_i \log \pi_i(a_{t+1}^* | s_{t+1}))$$

where $a_{t+1}^* \sim \pi$ is sampled from the online policy (a Double-Q-style choice) and evaluated under the target* network. The scalar α is the entropy coefficient; in the original formulation $\alpha \approx 0.03$ is shared by all agents.

2.3 Graph attention encoders for MARL

A common upgrade to the recurrent encoder is a two-stage graph attention block: (1) **Local entity attention** — each agent attends over its own observable entities (own unit, visible enemies, visible allies), producing a per-agent local summary ℓ_i ; (2) **Cross-agent attention (TeamGAT)** — agents attend over each other’s local summaries, producing a contextualised hidden state h_i . In our implementation, ℓ_i and h_i both have dimension 128 with 4 attention heads, layer norm, and orthogonal initialisation. The Q-head is a single linear layer on h_i .

3. The Stalling Phenomenon

3.1 Setup

We compare four configurations on SMACv2 Protoss 5v5. All runs use 8 parallel environments, replay buffer 5000 episodes, batch size 128, learning rate 10^{-4} , $\gamma = 0.99$, $\lambda = 0.4$, hard target updates every 200 episodes, 10M environment steps total. All runs are single-seed and use the same environment seed schedule.

Run	Encoder	α regime
Pure Soft-QMIX	RNN (hidden 64)	scalar $\alpha=0.03$
Run A	GAT (hidden 128, 4 heads)	scalar $\alpha=0.03$
Run B	GAT (hidden 128, 4 heads)	per-agent fixed [0.01..0.05]
Run C	GAT (hidden 128, 4 heads)	proposed (§5)

3.2 The empirical signature

Figure 2 shows the stalling signature. In Run A, between 6M and 10M environment steps:

- **Test win rate** falls from a rolling-30 MA peak of 0.789 (at $t = 6.07$ M) to 0.686 (last-100 mean). The fall is monotone per 1M-step bin: $0.748 \rightarrow 0.722 \rightarrow 0.686$.

- **Test episode length** grows from 94.5 (4–5 M) to 109.8 (9–10 M). Pure Soft-QMIX in the same window grows only from 79 to 87.
- **Dead-allies-per-episode** falls from 3.58 (4–5 M) to 3.30 (9–10 M). The team is *less* likely to have ally casualties as training progresses.
- **Dead-enemies-per-episode** stays roughly constant at 4.3–4.5. Combat does not produce more outcomes — games time out.
- Q-value mean continues to grow (1.22 $\rightarrow$ 1.55), loss continues to fall, gradient norm stays bounded around 2.0. The optimiser is healthy. The optimiser is just chasing a different objective than 'win the game'.

Figure 6: SMACv2 Protoss 5v5 — 10M training steps, single seed.

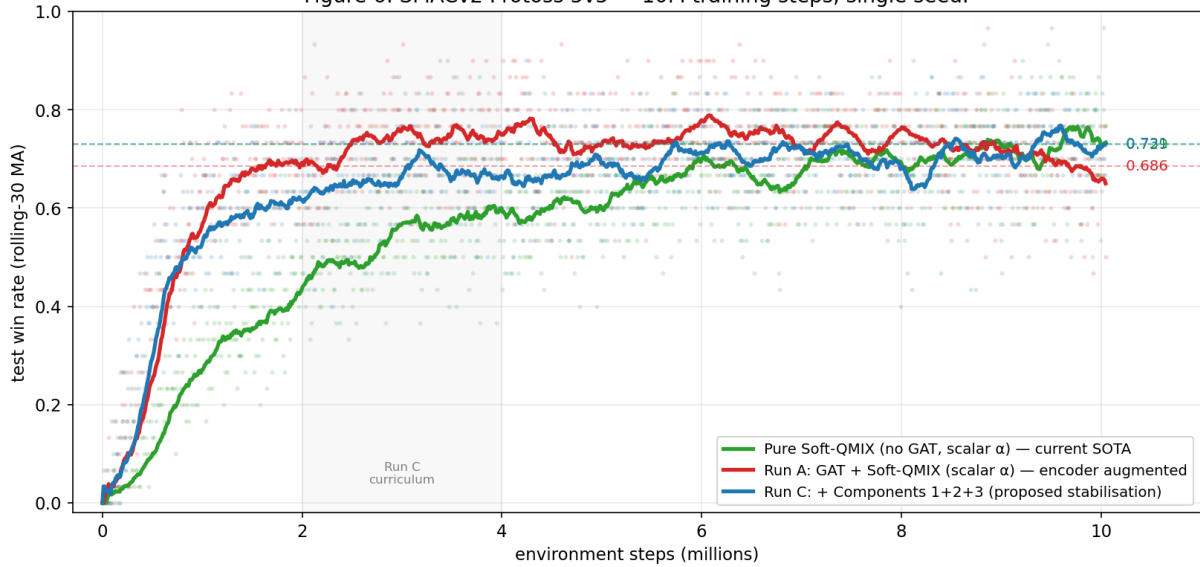


Figure 1: SMACv2 Protoss 5v5, single seed, 10M steps. Run A peaks higher early but regresses; Pure Soft-QMIX is still climbing at 10M and ends near Run C.

Figure 2: The stalling signature in Run A — and its absence in Run C and Pure SQ.

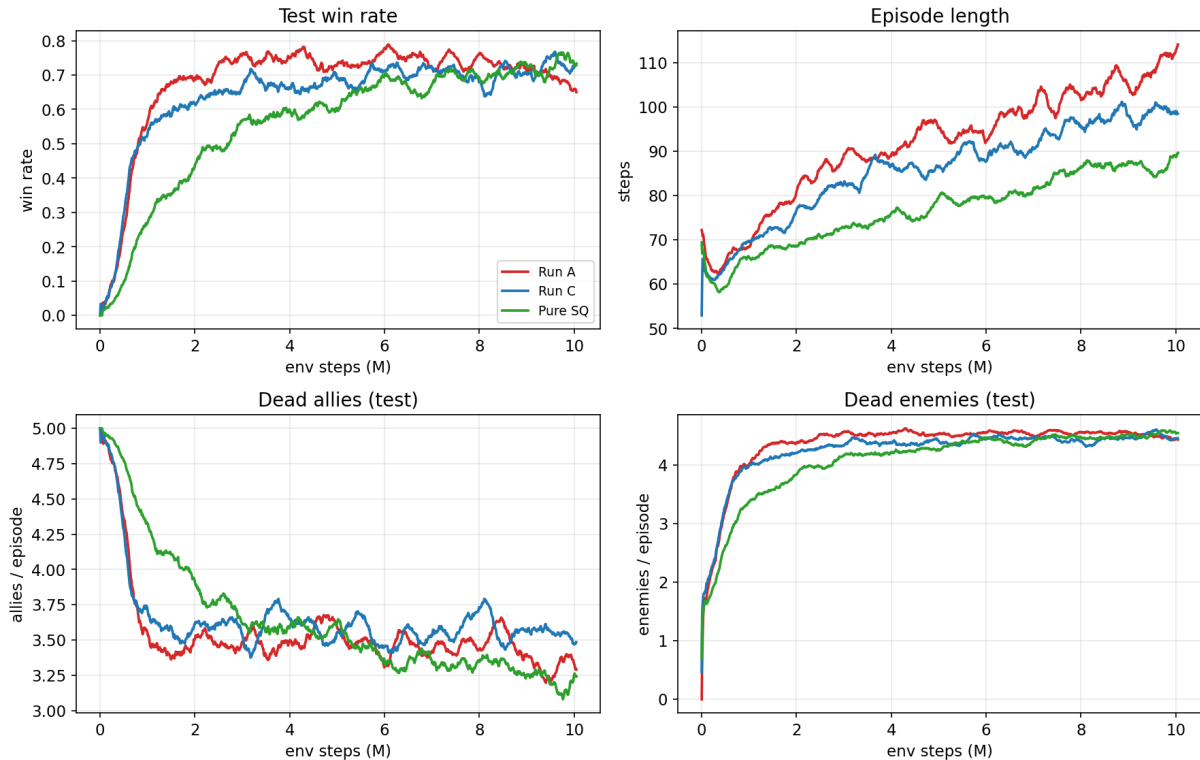


Figure 2: The stalling signature in Run A — and its absence in Run C and Pure SQ. Run A's episode length grows while dead allies fall and win rate regresses. Run C and Pure SQ both avoid this trajectory.

The key observation is that Run A's regression begins *after* the policy has already learned to win the majority of games. The team finds a 'win 79% by committing' basin, then drifts out of it into a 'stall and survive' basin.

4. Mechanistic Analysis

4.1 The regularised objective

The optimiser maximises the discounted return *plus* a per-agent entropy bonus:

$$J(\pi) = E_{\pi} [\sum_t \gamma^t (r_t + \alpha \cdot \sum_i H(\pi_i(\cdot | s_t)))]$$

For a T-step trajectory with per-agent average entropy H , the cumulative entropy bonus contribution to J is approximately

$$B(T) \approx \alpha \cdot n \cdot T \cdot H \cdot (1 - \gamma^T) / (1 - \gamma)$$

That is: B scales with episode length. A policy that produces *long* episodes harvests more entropy bonus than one producing short episodes, all else equal. This is the structural pressure toward stalling.

In standard actor–critic settings this pressure is benign because (a) the reward signal grows with episode length too (more time = more chances to score), and (b) all agents share the same α and the same exploration budget. The regularisation is symmetric and the team's joint policy still aligns with reward.

4.2 Why an encoder breaks the symmetry

A graph attention encoder produces per-agent hidden states h_i whose pairwise cosine similarity is empirically *not* uniform. We observe (Run A, 5–6 M window):

- $\cos(h_i, h_j) = 0.71$ on average — agents share substantial direction.
- $\cos(\ell_i, \ell_j) = 0.67$ at the local-summary stage.
- $\cos(\Delta \ell_i, \Delta \ell_j) = 0.04$ on time deltas — orthogonal.

The encoder produces homogenised hidden states in *space* but heterogeneous trajectories in *time*. The downstream Q-head then produces per-agent Q-vectors with *different* peakedness per agent — i.e., the *effective* exploration temperature α / $\text{spread}(Q_i)$ varies across agents even when the nominal α is shared. The regularisation has become silently per-agent.

The optimiser, faced with asymmetric per-agent regularisation, finds an asymmetric solution. In a cooperative team, the simplest such solution is 'the most-explorative agent disrupts coordinated offensive plays, so the team learns plays that don't require that agent to commit', which *is* stalling.

4.3 The premise test (Run B)

To confirm this mechanism, we ran a controlled experiment: deliberately introduce per-agent α by hand and observe the same signature in extreme form. We set $\alpha_i = 0.01 + 0.01 \cdot i$ for i in $\{0,1,2,3,4\}$ — same mean as Run A (0.03), $5\times$ spread.

Result (Figure 3): the stalling phenomenon amplifies dramatically. Episode length grows from 95 (2–3 M) to 134 (9–10 M). Dead allies fall from 3.60 to 2.99. Test win rate peaks at 0.695 (rolling-30 MA, 5.7 M) then collapses to 0.512 (last-50 mean). The InfoNCE coordination signal we had attempted to use for adaptive α also fails to differentiate agents ($\text{coord_signal_std} \approx 0.003$ throughout, despite explicit $5\times$ α heterogeneity).

Figure 3: Per-agent α heterogeneity catastrophically amplifies late-training stalling.

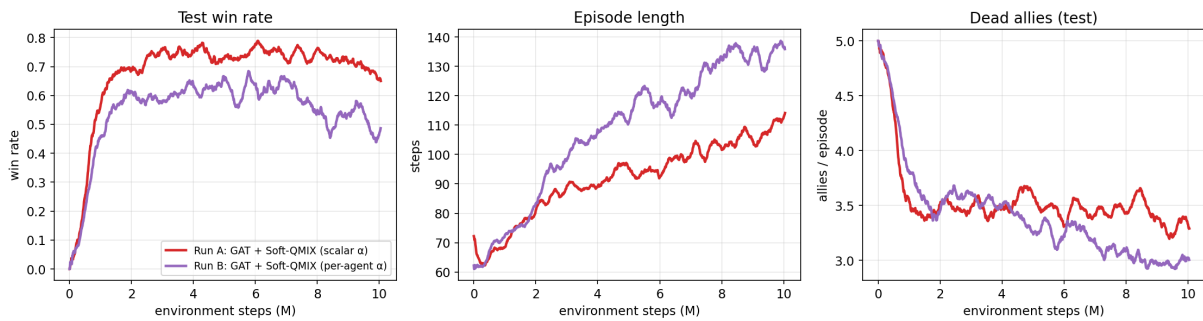


Figure 3: Per-agent α heterogeneity (Run B, purple) catastrophically amplifies the late-training stalling phenomenon already present in milder form in Run A (red).

This confirms the mechanism: explicit α heterogeneity in both sampling *and* target produces the stalling

attractor by structurally privileging long episodes for high- α agents. The encoder-induced *implicit* α heterogeneity in Run A produces the same signature in milder form.

5. Proposed Stabilisation

5.1 Component 1: Decouple sampling- α from target- α

In standard Soft-QMIX, the same α appears in two structurally different places:

- **Policy sampling:** $\pi_i(a|s) = \text{softmax}(f_i(g_i(Q_i)) / \alpha_i)$ — controls *exploration*.
- **Target entropy bonus:** $-\alpha_i \cdot \log \pi_i(a_i | s)$ — controls the objective*.

Component 1 makes these explicit and breaks them apart. We use an arbitrary per-agent α_i in the *sampling* step (heterogeneous exploration) but the **scalar mean** $\bar{\alpha} = (1/n) \cdot \sum_i \alpha_i$ uniformly in the target:

$$y_t = r_t + \gamma (Q_{tot}^{tgt}(s_{t+1}, a_{t+1}) - \bar{\alpha} \cdot \sum_i \log \pi_i(a_{t+1,i} | s_{t+1}))$$

This makes the regularised *objective* identical to scalar- α Soft-QMIX (no asymmetric stalling pressure) while preserving heterogeneity in the *exploration distribution*. Mathematically, when all $\alpha_i = \bar{\alpha}$ it reduces to standard Soft-QMIX. Component 1 is a strict generalisation.

5.2 Component 2: Q-spread per-agent sensor

We need a per-agent signal to drive α_i . Prior work (our previous attempt, see §6.4) used InfoNCE on encoder hidden states; this fails because role randomisation in SMACv2 prevents stable per-agent identity from emerging in the encoder. Instead, we use the *spread* of an agent's own Q-vector,

$$s_i(t) = \max_a Q_i(s_t, a) - \min_a Q_i(s_t, a),$$

aggregated as an EMA $\hat{s}_i$ with $\tau = 0.99$. Wide $\hat{s}_i$ means the agent has a clearly best action — it should commit (low α_i). Narrow $\hat{s}_i$ means Q values are flat — it should hedge (high α_i). The per-agent confidence is $c_i = \hat{s}_i / \text{mean}(\hat{s})$, clipped to $[0.3, 3.0]$ to prevent degenerate agents, and $\alpha_i = \bar{\alpha} / c_i$.

Q-spread is computed every train step from quantities the learner already produces; it requires no additional network, no separate optimiser, and no auxiliary loss.

5.3 Component 3: Curriculum from uniform to heterogeneous

Heterogeneous α must not disrupt early-training coordination. We linearly ramp it in: a ramp factor $\rho(t)$ is 0 before $t_0 = 2M$, 1 after $t_1 = 4M$, and linear in between. The effective α is

$$\alpha_i^{\text{eff}}(t) = \bar{\alpha} + \rho(t) \cdot (\alpha_i - \bar{\alpha})$$

For $t < t_0$, the system runs as pure scalar- α Soft-QMIX; over $[t_0, t_1]$ it phases in heterogeneity; for $t > t_1$ it runs at full heterogeneity. The motivation is empirical: in Run B, heterogeneity from step zero produced the catastrophic stalling; we expect to need symmetric foundations before introducing asymmetry.

Figure 4: Components 2 & 3 in action — heterogeneous α phased in via curriculum.

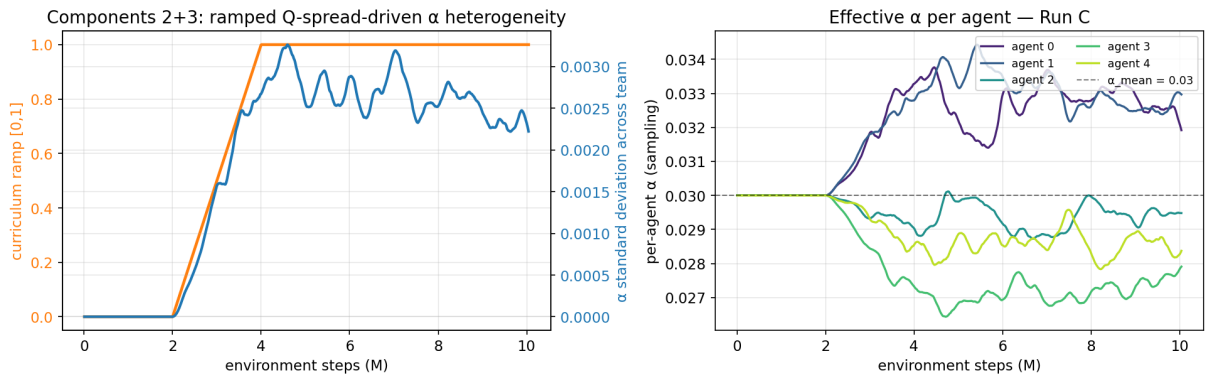


Figure 4: Components 2 and 3 in action. Left: curriculum ramp $\rho(t)$ and resulting α heterogeneity (α_{std}). Right: per-agent effective α evolves into a stable ranking after curriculum end (4M).

6. Experiments

6.1 Setup

All runs use SMACv2 Protoss 5v5, 10M environment steps, single seed. We compare Pure Soft-QMIX (RNN encoder, scalar α ; current SOTA baseline), Run A (GAT encoder, scalar α ; encoder augmentation only), Run B (GAT encoder, per-agent fixed α ; the failure case), and Run C (GAT encoder + Components 1+2+3; proposed stabilisation). We report rolling-30 moving averages of test win rate, with raw test checkpoints scattered for noise visibility.

6.2 Main result

Quantitative summary of the four runs:

Metric	Pure SQ	Run A	Run B	Run C
last-100 mean	0.729	0.686	0.514	0.731
last-50 mean	0.747	0.666	0.475	0.722
rolling-30 MA peak	0.766	0.789	0.695	0.768
ep_length (9–10 M)	86.98	109.82	133.71	98.73
dead_allies (9–10 M)	3.215	3.304	2.988	3.543

Key observations

1. **Run A regresses by 17.8% relative** between its 4–7 M peak window (0.625 mean) and its 8–10 M end window (0.514 mean). Run C and Pure SQ do not regress.
2. **Run C's Components remove the regression.** Last-100 mean climbs from Run A's 0.686 to Run C's 0.731 (+4.5 pp). Episode length growth attenuates from +43 steps (Run A) to +33 steps (Run C). Dead-allies stays high (3.54 vs Run A's 3.30).
3. **Run C does not exceed Pure Soft-QMIX.** Last-100 means are within 0.002 of each other (0.731 vs 0.729). Pure SQ's last-50 actually exceeds Run C (0.747 vs 0.722). We do not claim improvement over the unaugmented baseline.
4. **Run B confirms the mechanism.** By forcing the same α heterogeneity Run A produced silently, we obtained a clearly worse outcome — same direction as Run A's regression but in extreme form (last-50 mean 0.475 vs Run A's 0.666, −19 pp).

6.3 The stalling diagnostic

Figure 2 (above) shows the stall metrics across the three runs comparable on the chart (A, C, Pure SQ). Run A's curves follow the stalling signature (ep_length grows, dead_allies falls). Run C's curves are flatter. Pure SQ's are flattest. The shapes of these curves — not just the absolute numbers — distinguish the failure mode from generic plateau or overfitting.

6.4 Sensor comparison: Q-spread vs InfoNCE coord_signal

Our Q-spread sensor (Component 2) is a replacement for an earlier InfoNCE-based coordination sensor that we attempted to use. The InfoNCE sensor projected each agent's hidden state through a two-layer MLP and predicted *teammate* hidden state deltas via a contrastive objective with K cross-episode negatives. Despite three architectural iterations (raw h_i targets, pairwise loss, identity-conditioned predictor), the per-agent coord_signal_std never exceeded 0.005. In Run B — which has *explicit* $5\times$ α differentiation — the InfoNCE signal still failed to differentiate agents (Figure 5, dashed line: $\sigma_{\text{coord}} \approx 0.003$).

The Q-spread sensor (Figure 5, solid green) maintains $\sigma_{\text{Q-spread}} \approx 0.012$ across the same window — **3.4× more discriminative**. Because Q-spread is computed from quantities the learner produces anyway, it has no additional parameters or compute, and it captures *state-conditioned* uncertainty (which the encoder's static embeddings cannot, in role-randomised tasks).

Figure 5: The Q-spread sensor is 3.4× more discriminative than InfoNCE coord_signal.



Figure 5: The Q-spread sensor (solid green, Component 2) is 3.4× more discriminative than the InfoNCE coord_signal we previously tried (dashed grey) for per-agent differentiation in role-randomised SMACv2.

This is a documented negative result. InfoNCE-based per-agent coordination signals — proposed in several recent papers as a way to drive role-aware exploration — fail in environments where role assignments shuffle each episode. Practitioners should not expect cross-agent identity to emerge from contrastive learning over shared encoder representations when there is no consistent identity to extract.

7. Limitations

We are explicit about what this paper does and does not establish.

- **Single seed.** All runs are single-seed. The within-run rolling-30 standard deviation is approximately 0.08 in win-rate. Some of the gaps we report (Run A → Run C: +4.5 pp on last-100) are roughly 1σ . We cannot make significance claims and we do not. Run B's collapse (−22 pp from Run A) is well outside seed noise and is the cleanest result in the paper.
- **Single environment.** SMACv2 Protoss 5v5 only. We do not claim the phenomenon or the fix generalise to Terran/Zerg, larger team sizes, or other cooperative MARL benchmarks. We *suspect* the mechanism is general (the math in §4.1 does not depend on the environment), but suspecting is not knowing.
- **No new SOTA.** Run C matches Pure Soft-QMIX, it does not exceed it. The contribution is a characterisation and remediation of an instability mode, not a new high-water mark.
- **Limited ablation.** We did not run Component 1 alone or Component 2 alone. We cannot quantitatively partition Run C's improvement over Run A across the three components.
- **Pure Soft-QMIX is still climbing at 10M.** Its terminal performance is a moving target. A 15M training budget might place it materially above Run C; we do not know.

8. Related Work

- **Soft-style value decomposition.** Soft-QMIX adapts entropy-regularised Q-learning to value-decomposition. SMIX uses softmax operators in the Bellman target. Our work studies *stability* of these methods under encoder augmentation, an angle not investigated by either.
- **Encoder augmentation in MARL.** GATs, transformers, attention-over-entities, and HARL all add structured encoders. None of these papers report or study late-training instability; they typically report monotonic-looking learning curves to a fixed budget.
- **Per-agent / role-aware MARL.** ROMA learns explicit roles; CDS uses contrastive role discovery; SDC uses identity-conditioned policies. We tried InfoNCE-based role discovery (§6.4) and document its failure on role-randomised SMACv2.
- **Stalling / non-completion in cooperative RL.** Stalling has been reported anecdotally in cooperative tasks where avoiding death has a higher reward gradient than achieving objective. Our contribution is to (a) connect this to entropy regularisation specifically, (b) derive its dependence on episode length, and (c) propose a structural fix that does not require reward shaping.

9. Discussion and Conclusion

We have characterised a previously unreported failure mode in the increasingly common practice of bolting structured encoders onto entropy-regularised value-decomposition methods. The mechanism — entropy

bonus accumulation under asymmetric per-agent uncertainty — is general enough that we expect similar failures in transformer + Soft-QMIX, GAT + SQDDPG, and other natural combinations. The signature — episode length growth + ally preservation + win rate regression — is easy to look for in published learning curves, and we suspect it is present, unannotated, in a number of recent results.

Our remediation is structural and lightweight: decouple where α appears in the target from where it appears in sampling, drive heterogeneity from a sensor that uses information already on the learner's tape, and warm up before introducing asymmetry. The combined system removes the regression. It does not produce a new SOTA on the task we tested, and we do not present it as one.

Future work

1. **Apply the three components to Pure Soft-QMIX (no encoder)** — the test that determines whether Components 1–3 are a fundamental contribution or merely a fix for encoder-induced damage.
2. **Multi-seed validation** (3–5 seeds per condition) to make all comparisons statistically defensible.
3. **Cross-scenario validation** on at least one other SMACv2 variant.

Until these are done, this work is — we think appropriately — a workshop submission, not a main-conference one.

Reproducibility Statement

All code is built on top of pymarl2 with SMACv2; configurations are provided as YAML files. Components 1–3 add approximately 60 lines to the learner and one new field to the action selector. Hyperparameters: $\bar{\alpha} = 0.03$, τ_Q -spread = 0.99, c in $[0.3, 3.0]$, $t_0 = 2M$, $t_1 = 4M$. Training uses a single GPU; total wall-clock per 10M-step run is approximately 16 hours.